

Table 1: The expected sojourn time for the core parallel computational phase from our three interpolation approximations as a function of the system load $\rho = \gamma/\mu^*$ and the heterogeneity ratio $r > 1$ in comparison against simulation, with the lowest error highlighted in bold. All three approximations are exact when $r = 1$. Each simulation entry is the grand mean over 5 independent replications of 20 million jobs each, with a warm-up period of 500 thousand jobs discarded per replication; the reported 95% confidence interval is the Student- t interval across the 5 replica means.

ρ	r	Simulation	$\mathbb{E}[T_{FJ}]$ in		$\mathbb{E}[T_{FJ}^{(0)}]$ in		$\mathbb{E}[T_{FJ}^{(1)}]$ in	
		$\mathbb{E}[T_{\text{sim}}] \pm 95\% \text{ CI}$	$\mathbb{E}[T]$	Err	$\mathbb{E}[T]$	Err	$\mathbb{E}[T]$	Err
0.4	2	1.814 ± 0.001	1.824	+0.55%	1.834	+1.06%	1.825	+0.57%
	4	1.702 ± 0.001	1.704	+0.10%	1.717	+0.85%	1.705	+0.14%
	8	1.676 ± 0.001	1.676	+0.00%	1.681	+0.29%	1.676	+0.01%
0.8	2	5.081 ± 0.019	5.101	+0.40%	5.168	+1.71%	5.151	+1.37%
	4	5.014 ± 0.019	5.016	+0.03%	5.050	+0.71%	5.026	+0.23%
	8	5.004 ± 0.019	5.003	-0.01%	5.014	+0.21%	5.005	+0.02%
0.9	2	10.050 ± 0.083	10.063	+0.12%	10.170	+1.19%	10.150	+0.99%
	4	10.011 ± 0.084	10.009	-0.03%	10.050	+0.39%	10.023	+0.11%
	8	10.006 ± 0.084	10.002	-0.04%	10.014	+0.08%	10.004	-0.02%
0.95	2	20.096 ± 0.295	20.036	-0.30%	20.174	+0.39%	20.153	+0.28%
	4	20.075 ± 0.296	20.005	-0.35%	20.050	-0.12%	20.021	-0.27%
	8	20.072 ± 0.296	20.001	-0.35%	20.014	-0.29%	20.003	-0.34%